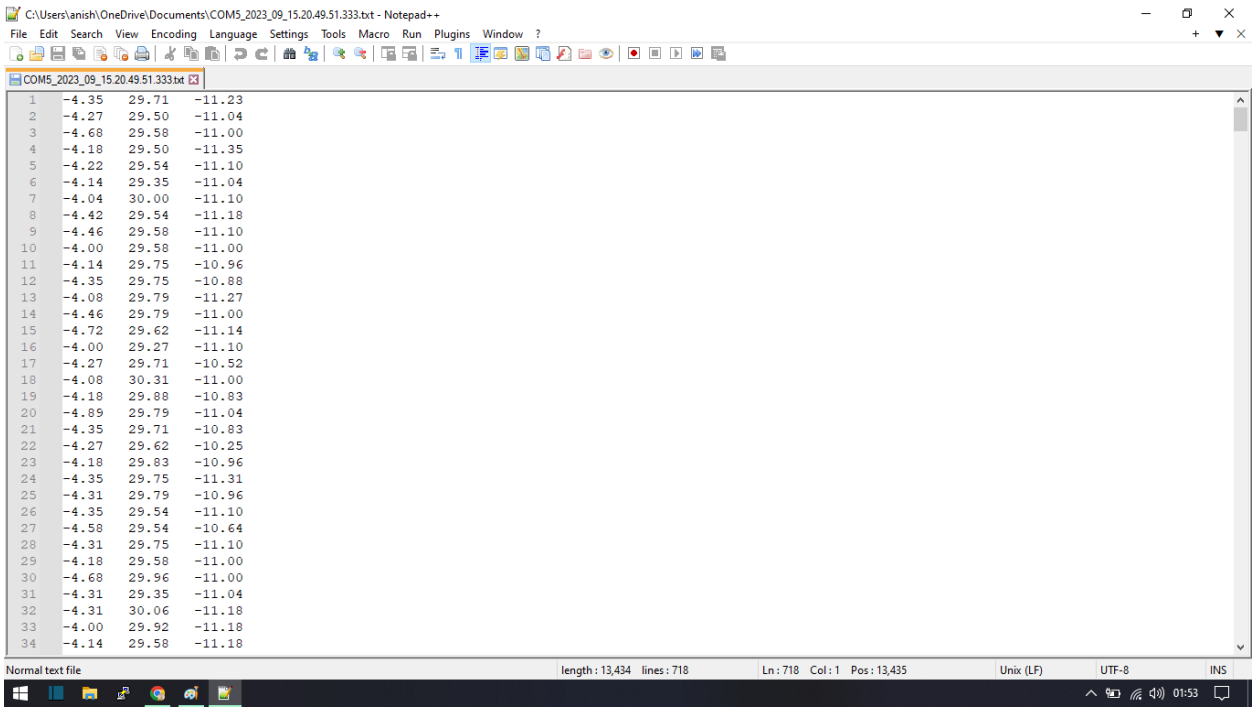
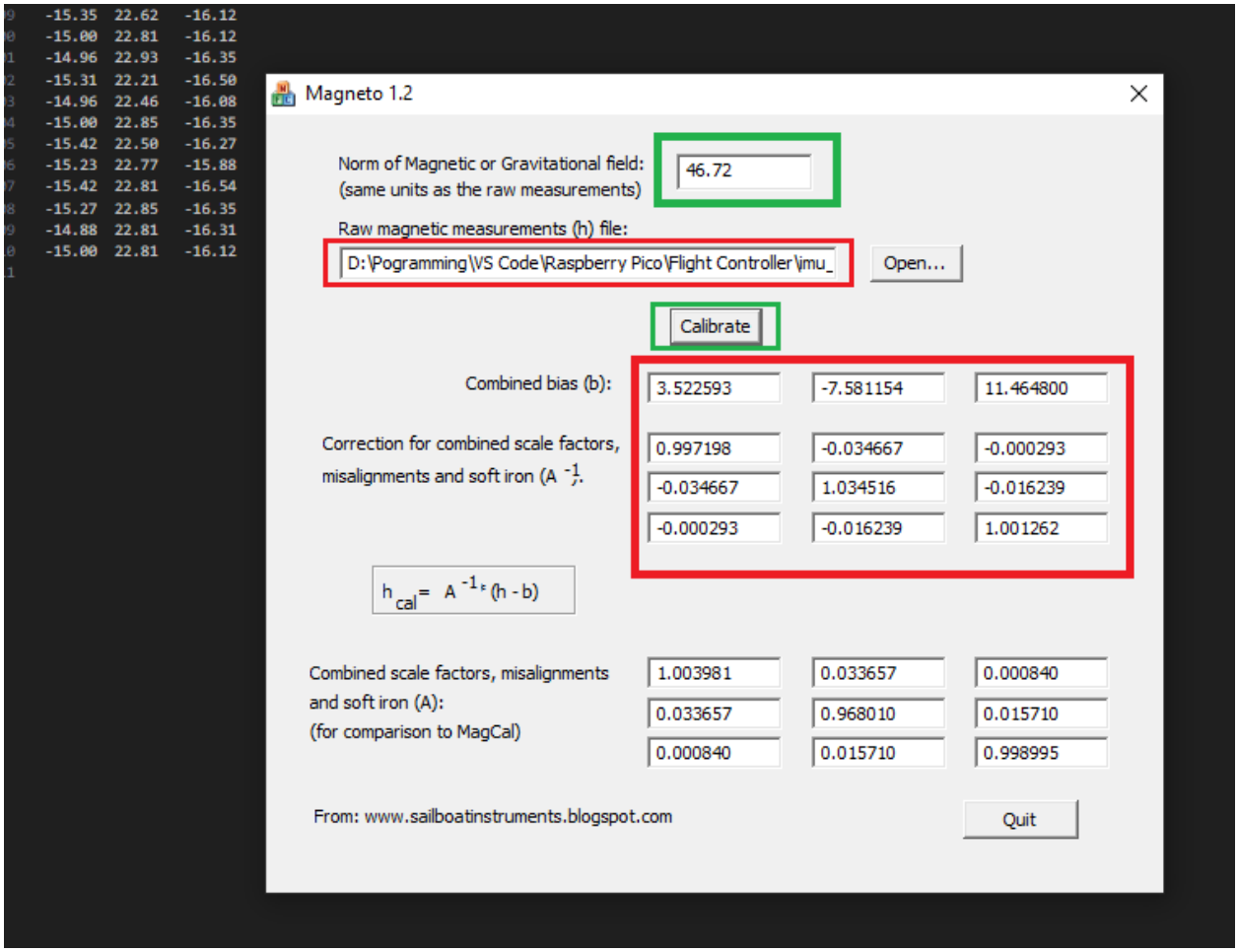
Compass Calibration Procedure

For the drone to be stable you need to calibrate the compass and it's mandatory. The calibration process is simple with some easy steps as noted below.

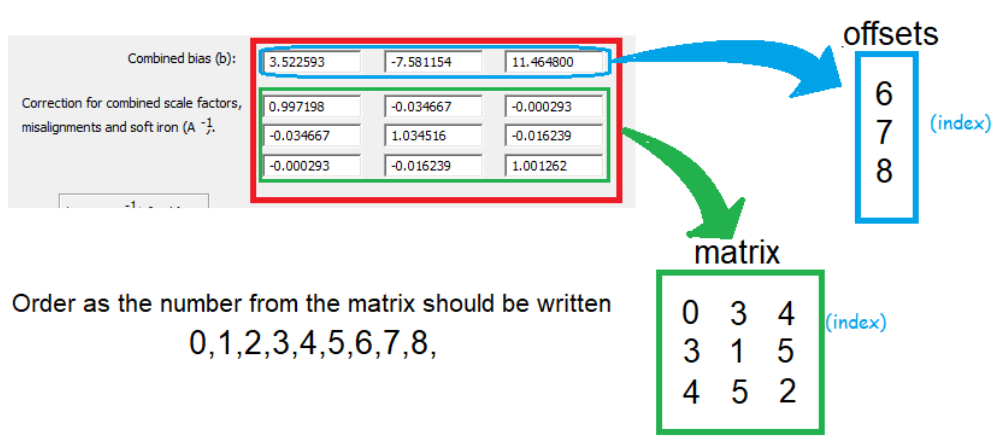
- Press & hold the bootsel button of the Pico and plug it into the pc via usb.
- Now drag and drop the compass_calibration.uf2 file into the pico drive. And the drive will vanish.
- Connect Pico to PC via a usb cable and open a serial monitor(you can use the arduino ide monitor) at 115200 baud rate. You will see columns of data printing. Now rotate the drone to each and every orientation possible. After you're done rotating, copy the whole data printout on the serial monitor and paste inside a normal text file inside the notepad. Now save it as a name, the file should be a txt file (example: calibration.txt)



- Open the magneto12_Calibration_Helper app and select the previously saved .txt file.
- Hit Calibrate and note the values of the matrix as shown in the images below.



- Now open the notepad and paste the values in proper order as shown in the image below.

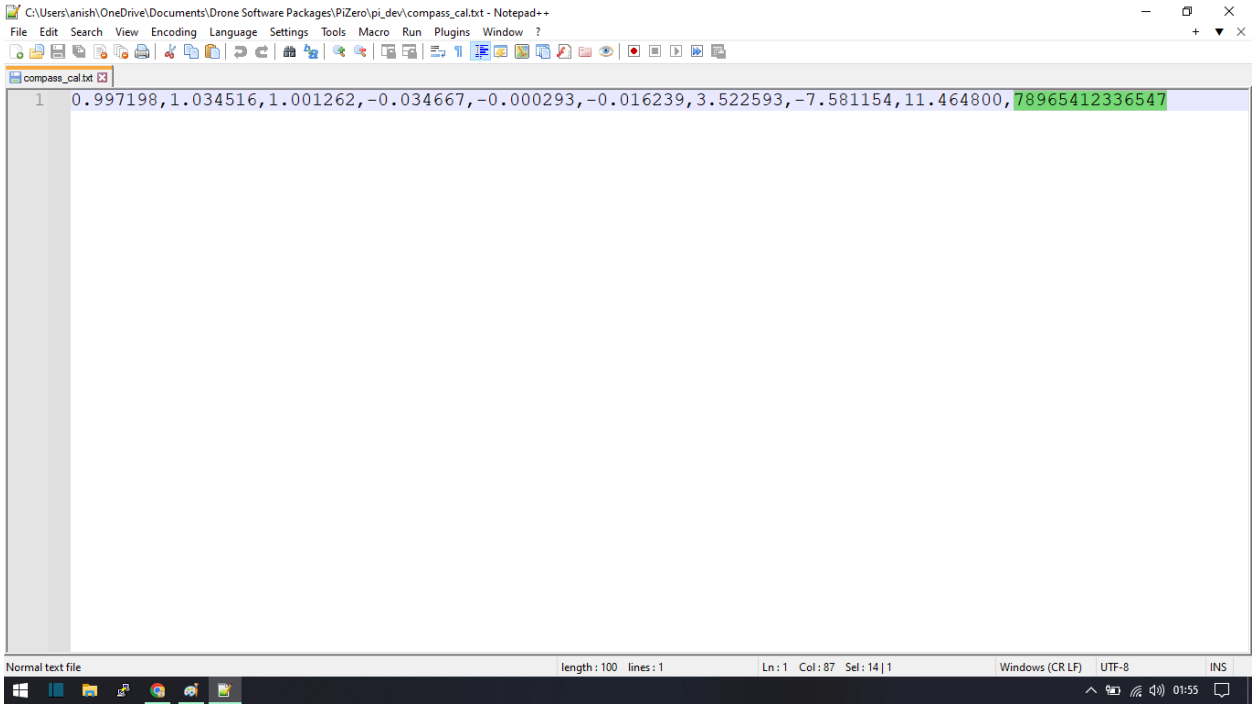


Order as the number from the matrix should be written

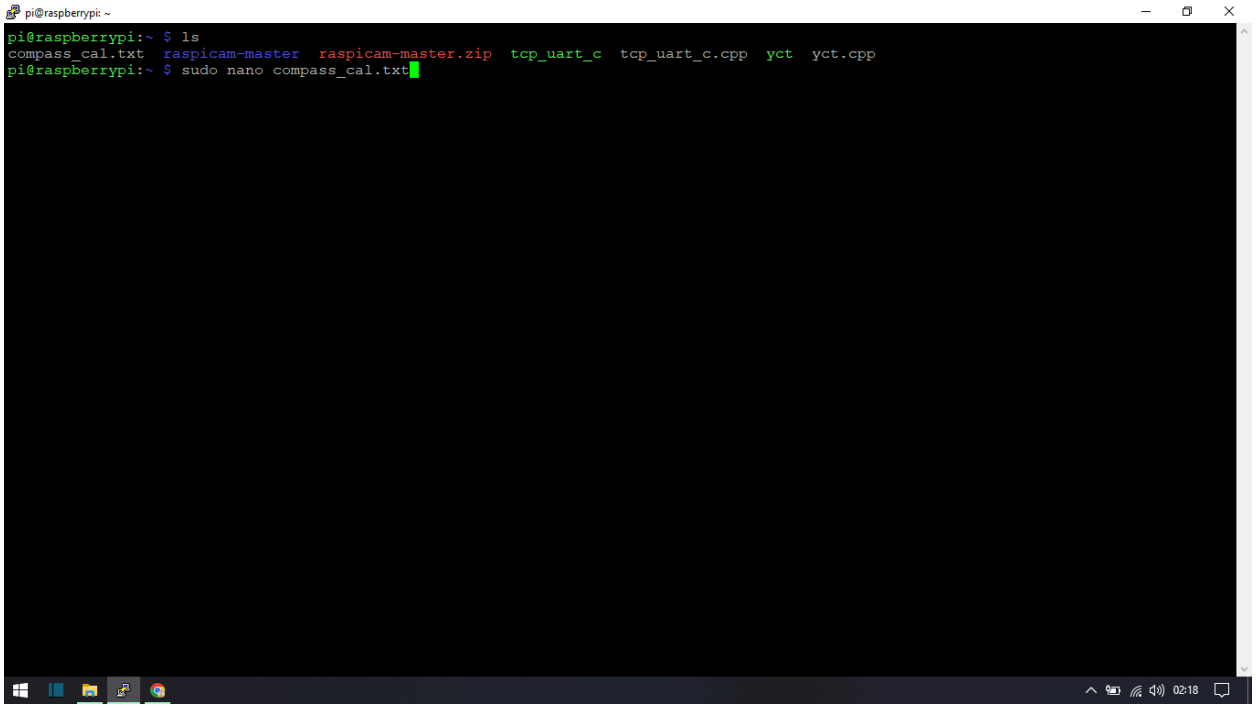
0,1,2,3,4,5,6,7,8,

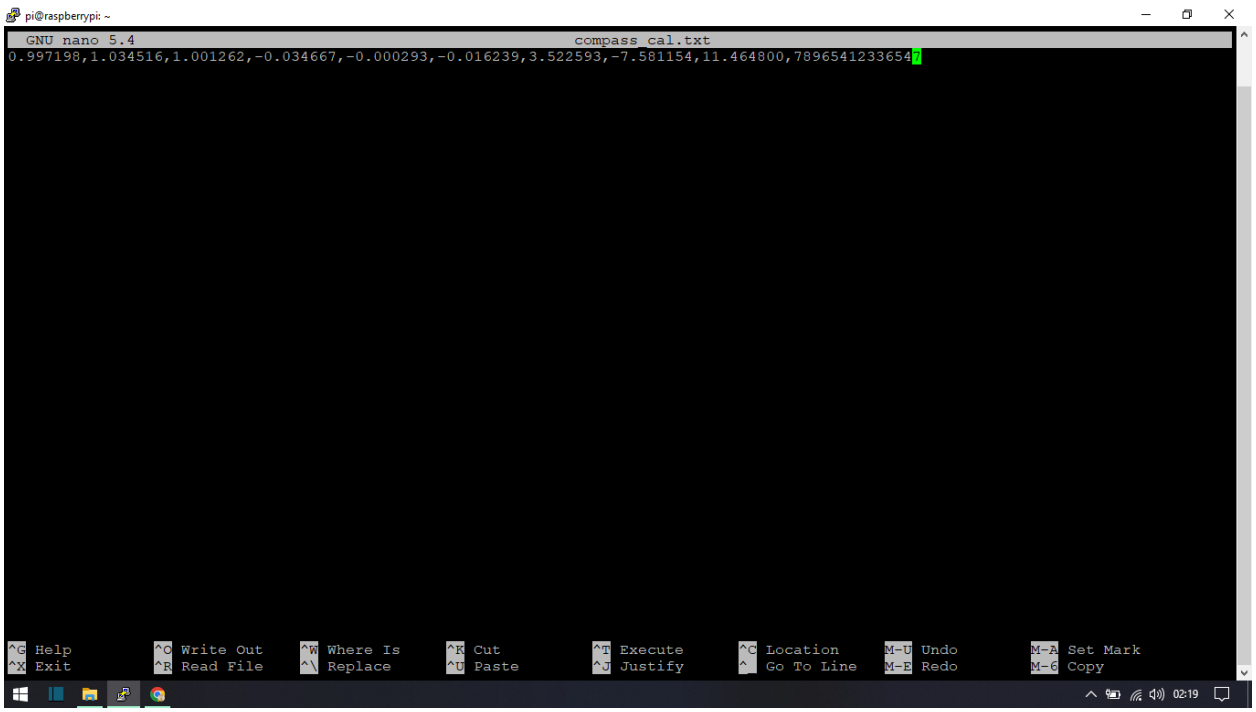
example : "0.997198,1.034516,1.001262,-0.034667,-0.000293,-0.016239,3.522593,-7.581154,11.464800,"

- But there's a catch, the text file should be in one line and 100 characters. So, after you paste the values the size will be less than 100. Now to make it 100 characters long fill with some random numbers. (green selection shown in the image below)

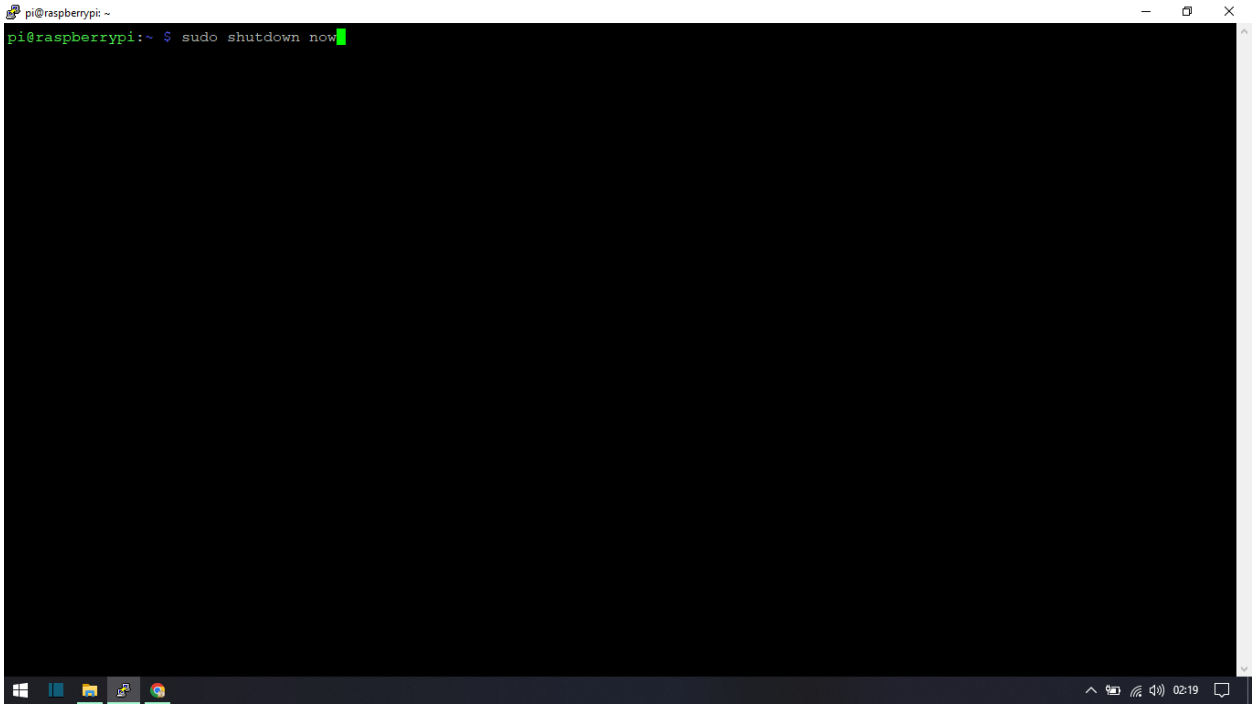


- Now copy this text and connect the pi zero to pc via usb and login to terminal and open the **compass_calibration.txt** file and erase previous values and type the new values one by one. Press **CTRL+S** then **CTRL+X** to save and exit.





- Shutdown the Pi zero with command sudo shutdown now



- Now disconnect usb and power the drone the led on the pico should be turned on. Wait a little and the led will start blinking, this means you have had a successful compass calibration. Now connect the drone to the app and shutdown the drone. After the led turns off in the raspberry pi zero turns off, disconnect the battery.

You have successfully completed the Compass Calibration Procedure. Return to the V4 Instruction PDF and follow next steps.